


React

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Lec 67 22 Jan 2024

→ JS library. React is flexible.

React banaya Meta ke developers ne.

It is basically used for creating user interface.

↓
frontend
design

React specialities :- ① it makes single page application (SPA).

single page application mei multiple pages hote hai but ek page se dusre page par jaane mei aisa lagta hai ki refresh/reloading nahi hoti.

MPA → multiple page application mei koi particular route hit hone par reloading hoti hai

→ alag alag route hit hone ke baad bhi page reload nahi hua.

SPA

- ① No reloading
- ② faster
- ③ Not good for SEO

MPA

- ① Reloading hoti hai
- ② Slower in comparison to SPA
- ③ good for SEO

SPA faster isliye hai kyunki server ki side se koi page banke nahi aata.

MPA less fast hai kyunki server se naya page banke aane mei time lagta hai.

SPA mei saara kaam client side par kiya jata hai isliye bhi faster hai. Same kaam DOM bhi karata hai.

Jo kaam React karega wo same kaam DOM se bhi kar sakte hai. But React kyu aaga picture mei?

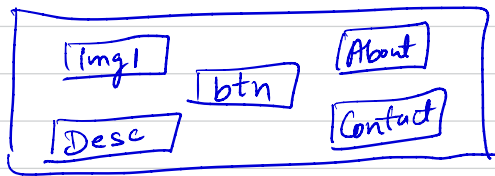
DOM

- ① It is operationally heavy. Matlab jyada bandwidth leegi which means jyada network calls lag rahi hai.

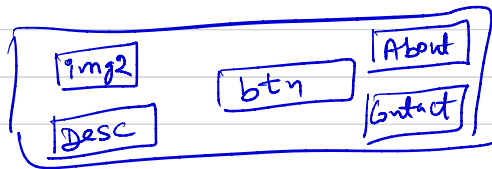
It is Not efficient to handle HTML.

React

Maan lo maine page 1 ke btn par click kiya, everything remains same except img 2/1. Ab DOM sab kuch fir se banayega/ load karega from starting to end.



page 1 (stage 1)



page 2 (stage 2)

img 1 → img 2 ban gayi but baaki sab same raha. Fir bhi sab kuch fir se load hua shuru se end tak jabki kaafi chize same thi. Isliye DOM is operationally heavy.

What is network calls? Browser ke network tab mei you can see network calls.

→ But React stage 1 and stage 2 ko compare karega and usi hisse ko banayega jo different hota hai, baaki sab ko create nahi karega from scratch kyunki vo same hai bho stage 1 & stage 2.

Efficiency improve ho gyi, time kam lagega.

VDOM → virtual dom

↓

Stage 1 and stage 2 mei comparison karta hai and sirf different hissa create karega and common/same chiz same rakhega; create nahi karega.

MPA → good for SEO (34:30)

SPA → not good for SEO

MPA mei page reloading hoti hai.

Browser mei web crawlers hote hai. ye web crawlers poori webpage par ek ek word dekhte hai and in words ke basis par indexing karte hai.

React kcase mei reloading nahi hoti different routes hit hone par; jiske karan Browser ko pata nahi chalega ki new page ~~na~~ aaya/bana hai. Ab web crawlers apna kaam kar nahi ~~ke~~ payenge jiske karan indexing nahi hogi. Hence react is bad for SEO. MPA mei new page par reloading hoti hai, hence it is good for SEO kyunki web crawlers har word ki indexing kar payenge.

React ka SEO acha nahi hota.

React → JS library

→ SPA → SEO better but faster hai

→ virtual dom

→ component based architecture

↓
component matlab portion of code

↓
you can divide your code in multiple portions. Isse ye fayeda hoga ki kisi ek code mei dikkat aane se dusre codes affect nahi hoga.

↓
portion of code

these portions of code can be reused.

React is a component based architecture and its React components can be reused by calling them when needed. (just like function).

So, react components are ^{actually} functions. ← ye aage explain kiya gaya hai.

Two types of components in React :-

- class based component → known as stateful compo.
- functional based component → " " stateless compo.

you can do dynamic changes with the help of stateful component but not with stateless components.

Stateless components / functional component ke madad se aap sirf static changes kar sakte ho.

Hooks are also functions.

Jab mai functional components ke saath hooks use karta hu, then I can use dynamic changes.

[Isliye react ke developers ne bola ki abhi hum class based components use karayenge nahi. Kyunki dynamic changes can be done with function based components + hooks and static changes can be done with function based component only.]

→ written in react documentation.
→ maine ek array banaya and use display karaya, it is a static change.

→ maine array banaya, then push kiya, then pop kiya, then use display karaya. It is dynamic change.

Static Change

- **Definition:** Static changes involve content or behavior that does not depend on user interaction or external factors to change after being delivered to the user.
- **Key Characteristics:**
 - Content is fixed once generated or deployed.
 - Changes require manual updates to the server or codebase, followed by redeployment.
 - Typically involves HTML, CSS, and static assets like images or videos.
- **Examples:**
 - A company's homepage with fixed text and images that only change when a developer updates the source files.
 - An "About Us" page with hard-coded team member details.

Dynamic Change

- **Definition:** Dynamic changes involve content or behavior that updates based on user interaction, database queries, or external APIs.
- **Key Characteristics:**
 - Content is generated **on-the-fly** using back-end logic or front-end scripting.
 - Changes can occur without redeployment, often leveraging JavaScript, AJAX, or frameworks like React, Angular, or Vue.js.
 - Relies on a back-end server (Node.js, Django, etc.) and/or a database (MongoDB, PostgreSQL, etc.).
- **Examples:**
 - A weather app that updates temperature and conditions dynamically when a city is selected.
 - User dashboards that display personalized content fetched from a database.
 - E-commerce websites where products and prices are fetched dynamically based on user filters.

What is life cycle method of class component?
(important for interview $\Rightarrow$ sirf theory padlo)

Both JQuery and React are JS library.

NextJS is framework of React and React is library of JS.

NextJS $\rightarrow$ server side rendering can be done
 $\rightarrow$ good for seo

Aapko React use karna aana chahiye to study and understand NextJS.

Refer this website legacy.reactjs.org to
study react.